

Elena Cossu, PhD

POLITICAL SCIENTIST — COMPARATIVE POLITICS & COMPUTATIONAL METHODS

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Political scientist working on comparative politics, political behaviour and the resilience of European democracies, with a methodological specialisation in causal inference and computational text analysis. My research studies democratic backsliding, populism and illiberalism, party competition, and the political attitudes of majorities and minorities across Europe, combining natural language processing (BERT, XLM-RoBERTa, large language models) with quasi-experimental and machine-learning approaches to causal inference (synthetic control, difference-in-differences, double machine learning). Peer-reviewed publications in outlets including Politics and Governance, with work under review at World Politics and a UCL Press book chapter; a decade of international, interdisciplinary collaboration across Sciences Po Paris, the European University Institute and UK government research; and teaching experience across undergraduate, master's and PhD levels, backed by a Max Weber Programme Teaching Certificate. PhD in Economics (quantitative political economy), 2024. Fluent English, working French, native Italian.

Employment

European Central Bank (ECB)

Frankfurt, Germany

DATA SCIENTIST

June 2026 - Present

- Applying machine learning, NLP and large language models to large-scale, multilingual text and structured data within a major policy institution
- Building reproducible, version-controlled pipelines (Python, Git)

European University Institute

Florence, Italy

MAX WEBER FELLOW

Sept. 2025 - June 2026

- Affiliated to the Robert Schuman Centre for Advanced Studies
- Developing NLP-based tools to measure and predict political and economic preferences across Europe
- Completed the Max Weber Programme Teaching Certificate, including guest teaching at the Freie Universität Berlin

SciencesPo Paris

Paris, France

POSTDOCTORAL RESEARCHER FOR THE AUTHLIB PROJECT

Apr. 2023 - Aug. 2025

- Affiliated to the Centre for European Studies and Comparative Politics (CEE)
- Data collection and modelling for the Horizon Europe project “Neo-authoritarianisms in Europe and the liberal democratic response” (AUTHLIB), funded by the European Commission
- Using Natural Language Processing (NLP) to map party ideological configurations, illiberalism and political cleavages across Europe
- Working under the direction of Prof. Jan Rovny and Prof. Caterina Froio at the CEE and Prof. Cointet at Medialab

National Centre for Social Research

London, United Kingdom

SENIOR POLICY EVALUATION RESEARCHER

Nov. 2021 - Oct. 2024

- Design and implementation of surveys and survey experiments to measure the effect of public policies
- Design and implementation of experiments (Randomised Control Trials) and quasi-experiments (Synthetic Control, Propensity Score Matching, Difference-in-Differences) to assess causality
- Clients included the UK Ministry of Justice, The National Lottery Community Fund, the UK Department for Education, the Education Endowment Foundation, and the Department for Health and Social Care (DHSC)

Corvinus University of Budapest

Budapest, Hungary

ASSISTANT LECTURER

Sep. 2021 - Jan. 2022

- Assistant teacher for Master-level courses

European Commission — FATIGUE ITN

Budapest and London

EARLY STAGE RESEARCHER (MSCA)

Aug. 2018 - Sep. 2021

- Researcher on the Horizon 2020 project FATIGUE, a Marie Skłodowska-Curie Innovative Training Network (MSCA) on illiberalism and transformational fatigue in Central and Eastern Europe
- Quantitative analysis and writing of reports and articles for long-term comparative research projects
- Duty stations: (1) European Bank for Reconstruction and Development (EBRD), Office of the Chief Economist; (2) UCL School of Slavonic and East European Studies (SSEES); (3) Corvinus University of Budapest

Education

Corvinus University and University College London

Budapest and London

PHD IN ECONOMICS (QUANTITATIVE POLITICAL ECONOMY)

Aug. 2018 - Apr. 2024

- Dissertation: 'Essays on Conceptualising Policies Linked to Populist Rhetoric in Europe and Measuring their Effects in Hungary'
- Co-supervised at University College London (UCL); degree conferred April 2024
- PhD sponsored by the European Commission Marie Skłodowska-Curie Innovative Training Network (MSCA) as part of the FATIGUE research project on illiberalism in Central and Eastern Europe

SciencesPo Paris and Bocconi University

Paris and Milan

DOUBLE MASTER'S DEGREE IN INTERNATIONAL ECONOMICS AND MANAGEMENT

Aug. 2014 - Apr. 2017

- MA in International Public Management with a specialisation in Global Risks from SciencesPo Paris School of International Affairs (PSIA); dissertation «Fighting terrorism and fundamental rights»
- MSc in Economics and Management of International and Public Affairs from Bocconi University; dissertation «The Parisian transport system: a best practice of urban mobility»

Università Cattolica del Sacro Cuore

Milan, Italy

BACHELOR'S DEGREE IN POLITICAL SCIENCE AND INTERNATIONAL RELATIONS

Sep. 2011 - Oct. 2014

- Final dissertation: «The Hungarian Transition from Planned to Capitalist Economy»
- Erasmus at Pázmány Péter Catholic University in Budapest during the 2013/2014 academic year

Periods of leave without research activity since the PhD: none (no career breaks).

Peer-Reviewed Publications

How Critical Events Shape Political Agendas on Multiculturalism in France

POLITICS AND GOVERNANCE

2025

- <https://www.cogitatiopress.com/politicsandgovernance/article/view/9743>
- Investigates how terror attacks influenced political discourse on multiculturalism in France between 2014 and 2021. Analysing 143,870 tweets from major political parties with an XLM-RoBERTa Natural Language Inference model, the study tracks how far-right discourse shaped mainstream party communication on immigration and cultural diversity.

Application of Natural Language Processing to the Electoral Manifestos of Parties Characterised by Populist Rhetoric in Central and Eastern Europe

INTERSECTIONS. EAST EUROPEAN JOURNAL OF SOCIETY AND POLITICS

2025

- <https://bit.ly/42TL65F>
- Applies a range of NLP techniques to the electoral manifestos of populist parties in Central and Eastern Europe, uncovering rhetorical patterns and ideological shifts and showing quantitatively how discourse mobilises cleavages around Europe and corruption.

Clustering and Analysing Relevant Policy Dimensions of Populist, Left-Wing, Centrist, and Right-Wing Parties across Europe

CENTRAL EUROPEAN JOURNAL OF PUBLIC POLICY, 17(1), 41–54

2023

- <https://doi.org/10.2478/cejpp-2023-0004>
- A configurational analysis of the policy positions of 242 European parties (2014 and 2019) using the Chapel Hill Expert Survey and PopuList data. Cluster analysis shows that, rather than a classical left–right divide, Europe splits into four groups: right-wing populists, a moderate pro-Europe left, a pro-Europe pro-liberalism centre, and an intersectional identity-based left.

The Economic Effect Of Populist Rhetoric In Hungary

ONLINE JOURNAL OF MODELLING THE NEW EUROPE, 41, 105–138

2023

- <https://doi.org/10.24193/OJMNE.2023.41.05>
- Estimates the effect of populist rhetoric on GDP in Hungary (2010–2023) using Difference-in-Differences, Synthetic Difference-in-Differences, and the Synthetic Control Method.

Understanding Contemporary Populism Through the Latin American Experience

AARMS – ACADEMIC AND APPLIED RESEARCH IN MILITARY AND PUBLIC MANAGEMENT SCIENCE, 20(3), ARTICLE 3

2021

- <https://doi.org/10.32565/aarms.2021.3.4>
- Draws on the Latin American experience to interrogate contemporary European populism, reviewing the evolution of the concept and why the discipline still lacks a shared definition.

The Gravity Model for Trade Theory

KÖZ-GAZDASÁG, 14(3), ARTICLE 3

2019

- <https://doi.org/10.14267/RETP2019.03.25>
- A theoretical overview of the gravity model, its applications, and its limitations in the analysis of trade flows.

Book Chapters

The Effect of Policies of Populist Parties in Power on Economic Performance: The Case of Hungary

Cossu, E.

2025 In J. Kubik & R. C. M. Mole (Eds.), *The Anatomy of Right-Wing Populism: Dealing with Transformational Fatigue in Central and Eastern Europe* (Chapter 6). London: UCL Press.

<https://uclpress.co.uk/book/the-anatomy-of-right-wing-populism/>

Under Review

Diversity and Democracy: Muslim Minorities and the Support for Democracy in Western Europe

Submitted to *World Politics*

Rovny, J., Cossu, E., & Sattelmayer, L.

2026 Can diversity and democracy coexist? Using European Social Survey data, the article pairs a transparent linear-regression baseline with EconML's Double Machine Learning estimators (LinearDML and DR Learner) and conditional inference trees to study the democratic support of Muslim minorities in western Europe. It shows that more secular or liberal Muslim minorities are significantly more likely to support democracy than the majority, suggesting that western European Muslims are, overall, an asset for liberal democracy.

Working Papers

A Framework for Labelling Complex Political Concepts Using Open-Weight LLMs: Evidence from Illiberalism in European Party Communication

Cossu, E.

2026 Develops a six-node decision framework for LLM-assisted annotation of complex, contested political concepts, guiding researchers from concept assessment through model selection, validation and deployment. Validated on illiberalism in European party communication using 1.5 million tweets from 55 parties across seven countries (2001-2024): open-weight LLMs with codebook-guided prompts achieve $\kappa = 0.71$ and $F1 = 0.83$ against human labels, and knowledge distillation preserves party rankings ($\rho = 0.87-0.94$). Identifies strongly differential model errors for euroscepticism at the party level, and offers a lightweight, explicit annotation pathway for researchers without GPU infrastructure or human coders.

Women and AI: A Large-Scale, Validated Mapping of Women's Under-Represented and Under-Recognized Contributions to AI Research

Binelli, C. & Cossu, E.

SocArXiv preprint. https://doi.org/10.31235/osf.io/cxm6_v1

2026 Examines gender representation in AI scholarship across 500,000+ papers (2023-2026). Women are underrepresented in AI authorship, especially in data science; their work skews toward applied, social and purpose-driven research in health, education and fairness; and despite equal citation within topics, women receive fewer citations overall because their work concentrates in less-cited subjects.

Works in Progress

Blurring the Boundaries: Conceptualizing and Measuring Illiberal and Far-Right Party Communication in Europe

Cossu, E., Froio, C. & Völker, T.

2026 Maps illiberal elements in the communication of mainstream European far-right parties using all tweets posted between 2006 and 2023 by their official and leaders' accounts. Using BERT, each tweet is classified along four dimensions — nativism, authoritarianism, populism, and references to liberal-democratic institutions (checks and balances, judicial independence, media freedom) — to estimate how far far-right communication also articulates illiberal discourse, and how this varies over time and across parties.

Teaching Experience

Populism and Illiberalism in Europe: Tracking Opinions and Measuring Consequences

2026

FREIE UNIVERSITÄT BERLIN (FU)

- Guest lecturer within Prof. Bruno Castanho Silva's MA course, and a parallel standalone session, delivered as part of the Max Weber Programme Teaching Certificate. Developed lesson plans, slide decks and in-class activities; collected and reflected on anonymous student feedback.

LLM Techniques, Text Analysis and Illiberal Speech (Methods Seminar)

2026, 2024

SCUOLA NORMALE SUPERIORE (SNS), PISA

- Hands-on methods workshop for PhD researchers on the full text-as-data pipeline: web scraping, pre-trained models, BERT-based classification and generative LLMs, applied to the detection of illiberal and anti-democratic discourse, with practical work in Python and Hugging Face.

Policy Analysis and Assessment Methods

2024

SCIENCESPO PARIS

- Master's course (up to 100 students) on the theory and practice of policy evaluation, from conceptual frameworks to machine-learning approaches, with applied examples in R and Python.

Democracy in Europe: Political Economy of Populism

2024, 2023

SCIENCESPO PARIS

- Master's seminar on democratic institutions and their resilience in Europe and on the drivers and effects of populist rhetoric, with case studies from EU member states.

Text as Quantitative Data (PhD Seminar)

2023

SCIENCESPO PARIS

- PhD seminar introducing computational text analysis and NLP techniques for political and social science research.

Impact Evaluation: Quasi-Experimental Design

2023, 2022

NATCEN FOR THE UK OFFICE FOR NATIONAL STATISTICS (ONS)

- Applied professional training on causal inference and quasi-experimental methods (DiD, RDD, IV, matching) for policy evaluation, delivered to UK government statisticians.

International Organizations & Economic Diplomacy; Global Economic Trends; Development Economics

2021, 2018

CORVINUS UNIVERSITY OF BUDAPEST

- Seminars on global economic governance, international organisations, and the main theoretical and empirical debates in development economics.

Selected Presentations and Conferences

ECPR Joint Sessions, 'From Code to Conversation: What LLMs Mean for Political Science'

- 2026 A Framework for Labelling Complex Political Concepts Using Open-Weight LLMs: Evidence from Illiberalism in Party Communication

International Political Science Association (IPSA) Conference

- 2025 How Critical Events Shape Political Agendas on Multiculturalism in France

European Political Science Association (EPSA) Conference

- 2025 The Social Bases of Democracy in Europe

CIVICA Political Behaviour and Institutions Conference (PoBI); ECPR Joint Sessions (Text-as-Data);

- 2025 HEPP5, Helsinki
Measuring Illiberalism Using Text Data and Machine Learning

MethodsNet Conference

- 2024 Measuring Illiberalism Using Text Data and Machine Learning

ECPR General Conference

- 2022 Assessing the Economic Performance of a Government Characterised by Populist Rhetoric in Hungary: A Synthetic Control Approach

WINIR Symposium; RSA Winter Conference; UACES 49th Annual Conference

- 2019 Economic Populism, Fiscal Policies and Budget Deficits in Central and Eastern Europe

Grants, Awards and Service

- 2023 ACRI Young Investigator Prize, ECOMOD (for contributions in applied computational/economic modelling)
- 2023 AUTHLIB — Neo-Authoritarianisms in Europe and the Liberal Democratic Response (EU / UK Research and Innovation)
- 2018 FATIGUE — Delayed Transformational Fatigue in Central and Eastern Europe, Horizon 2020 (Grant ID: 765224)
- 2023 Blind peer-reviewer, East European Politics and Societies

Relevant Skills

Causal inference & quantitative methods

- Randomised Control Trials (RCTs) and survey experiments
- Difference-in-Differences and Synthetic Difference-in-Differences
- Synthetic Control Methods (SCM)
- Matching and Propensity Score Matching/Weighting (PSM/PSW)
- Double Machine Learning (DML / EconML)
- Cluster analysis, imputation, survey design

Computational text analysis

- Natural Language Processing (NLP); BERT and XLM-RoBERTa fine-tuning
- Large language models (open-weight LLMs) for labelling and evaluation
- Multilingual sentiment/stance analysis; web scraping

Programming: R, Python, SQL, Git, STATA, SPSS, HTML, LaTeX

Languages: English (fluent), French (working), Italian (native), Spanish (conversational), Hungarian (basic)

Certifications

- 2026 **European University Institute** Teaching Certificate (Max Weber Programme)
- 2023 **Google** Data Analytics Professional Certificate
<https://www.coursera.org/account/accomplishments/verify/XZCBSTWMZ3W4>
- 2022 **Office for National Statistics (UK)** Accredited Researcher
- 2019 **Centre for Microdata Methods and Practice (cemmap)** Policy Evaluation Methods

References

- Professor Jan Rovny, Sciences Po Paris, Centre for European Studies and Comparative Politics (CEE), jan.rovny@sciencespo.fr
- Professor Caterina Froio, Sciences Po Paris, Centre for European Studies and Comparative Politics (CEE), caterina.froio@sciencespo.fr